



Semester 1 Examinations 2019 / 2020

Exam Code(s)	4BCT1, 4BSE1, 1MSME1, 1MEEE1, 1MECS1, 1CSD1, 1CSD2, 2SPE1, 1MAI1
Exam(s)	B.Sc. Degree (Computer Science and Information Technology) B.E. Energy Systems Engineering MSc in Mechanical Engineering ME (Electrical and Computer Engineering) ME (Electrical and Electronic Engineering) M.Sc. in Computer Science (Data Analytics) M.Sc. in Computer Science (Artificial Intelligence)
Module Code(s)	CT561
Module(s)	Systems Modelling and Simulation
Paper No.	I
External Examiner(s)	Dr. Jacob Howe
Internal Examiner(s)	Prof. Michael Madden *Dr. Jim Duggan

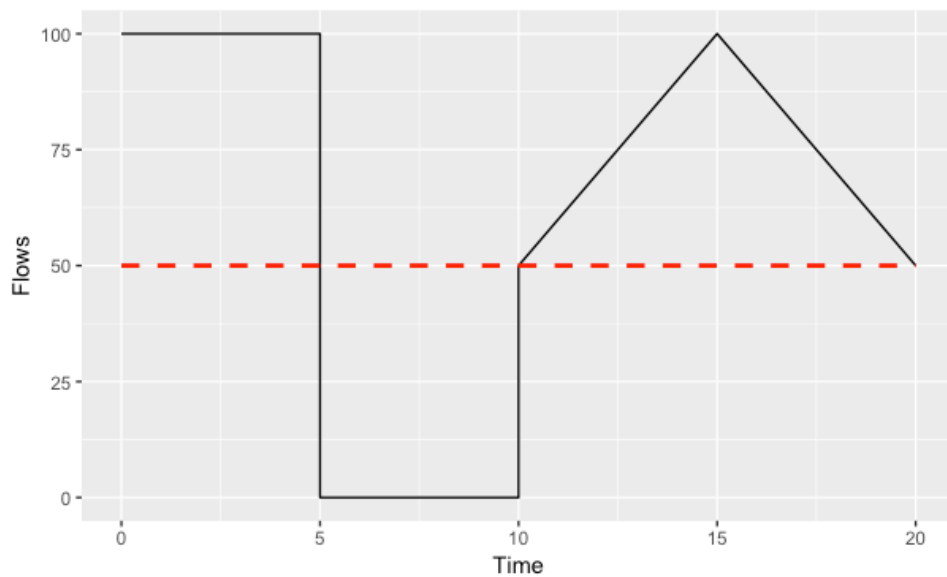
Instructions:

Answer any 3 questions. All questions carry equal marks.

Duration	2hrs
No. of Pages	5 (Including Cover Page)
Department(s)	School of Computer Science
Requirements	Graph Paper

1. (a) Based on the flow variables shown in the diagram (dashed line is the outflow, solid line the inflow), and assuming that the initial value of the stock is 100, perform the following:

- Draw a stock and flow model (including the net flow).
- Calculate, and show, the net flow.
- Use graphical integration to plot the stock over time, clearly showing the different behaviour modes for each time interval



[12]

- (b) Describe Euler's method of integration, and state the assumptions.

[5]

- (c) Calculate the stock using Euler's equation where $DT=5$. Estimate the integration error for each time step of 5.

[8]

2. (a) Construct a stock and flow model from the following description of a service industry company. Specify the stock equation (there is no need to document equations for flows or auxiliaries), and clearly show the link polarities, and feedbacks, on the diagram.

- **Customer Orders** increases the **Service Backlog (Stock)**, while **Order Fulfillment** decreases the Service Backlog
- An increase in **Service Backlog** leads to an increase in **Desired Service Capacity**.
- An increase in **Service Standard** leads to an increase in **Desired Service Capacity**
- An increase in **Desired Service Capacity** leads to an increase in **Work Pressure**.
- An increase in **Work Pressure** leads to: (1) a reduction in **Time Per Order**, and (2) an increase in **Work Intensity**.
- A reduction in **Time Per Order** leads to a reduction in **Service Standard**. A fall in **Service Standard** leads to a corresponding decline in **Time Per Order**.
- An increase in **Time Per Order** leads to a reduction in **Order Fulfillment**.
- An increase in **Work Intensity** leads to (1) an increase in **Order Fulfillment** and (2) an increase in **Fatigue**.
- An increase in **Fatigue** leads to a decrease in **Order Fulfillment**.

[12]

- (b) Based on the stock and flow model in part(a):

- Show the following feedback loops on the model: *“Goal Erosion”*, *“Burnout”*, *“Corner Cutting”*, and *“Overtime”*.
- Starting with the proposition that the Customer Orders are increasing, confirm the polarity for each of these four loops.

[8]

- (c) What are the advantages of building this feedback model? How can its insights be use to identify ways to improve the Service Industry?

[5]

3. (a) Describe the stock and flow structure (with equations) of the following model structures, and show their expected behaviour over time.

- Fractional Increase Rate
- Fractional Decrease Rate
- Adjustment to a Goal

[4]

- (b) Consider the following description of a University model. Construct a stock and flow model, showing the link polarities and the feedback loops (with polarities).

- *Enrollments* (Enrollment Fraction = 0.25) increase the stock of **Students** (10000), while *Graduations* (Graduation Fraction = 0.23) and *Attritions* (Reference Attrition Fraction = 0.02) reduce the student stock.
- The *Desired Number of Staff* is the ratio of **Students** to *Desired Student Staff Ratio* (initially 20).
- *Change in Staff* (Net Flow) increases/reduces **Staff** (500), based on an adjustment of the *Discrepancy in Staff* (AT=2).
- The *Student to Staff Ratio* is based on the ratio of **Students** to **Staff**.
- The *Attrition Fraction* is formulated as an effects structure that depends on the *Student Staff Ratio*. Assume the *Reference Student Staff Ratio* = 0.02, and that the *Effect of Student Staff Ratio on the Attrition Fraction* has a minimum value of 0.01, and a maximum value of 0.10.

[8]

- (c) Specify the full set of stock and flow equations, including the effects structure (the effect relationship should be shown on an x-y plot).

[8]

- (d) Assume that the model starts in equilibrium in 2019, and runs to 2030. At time 2024, the *Desired Student Staff Ratio* increases to 25. Show how the *Effect of Student Staff Ratio on Attrition Fraction* will change as a result of this increase.

[5]

4. (a) Give a definition of reproduction number R_0 , and describe why it is an important measurement to help control the spread of infectious diseases. What is the relationship between R_0 and the goal of reaching *herd immunity* in a population? Show a practical example, where an influenza outbreak is assumed to have $R_0=2$.

[5]

- (b) Show the Stocks, Flows and Feedbacks for the SIR Model. Assume the following values.

- $S_0 = 9999, I_0=1, R_0=0$
- Contact Rate = 6/day
- Infectivity = 0.25
- Recovery delay = 2 days

Extend the model by including the possibility of vaccinations, where vaccines are a finite resource, and that the average vaccination delay is 2. If there are no vaccines, no vaccinations can take place.

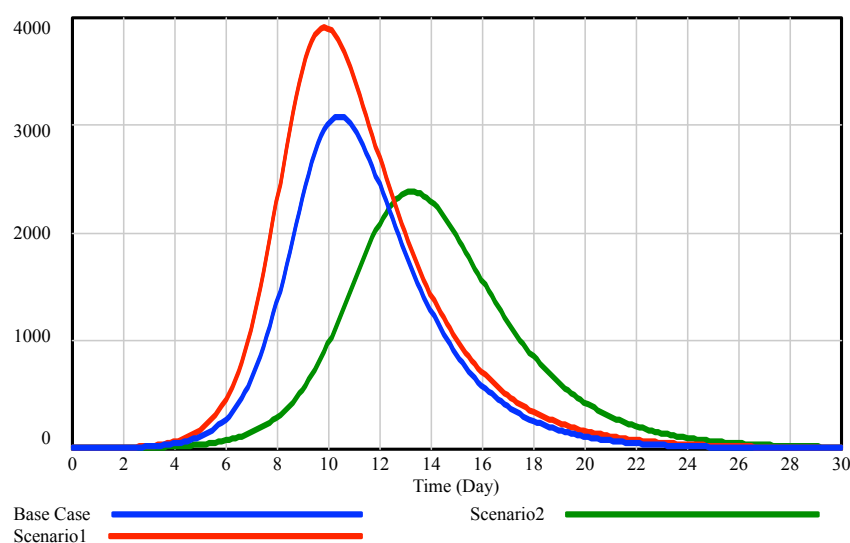
[12]

- (c) Discuss how the parameters *vaccination delay*, *contact rate* and *infectivity* can impact the infected curve. How might public health policies be formulated to impact these three parameters?

[4]

- (d) Based on the following graph, where the *Base Case* (blue) is the middle curve, and the highest peak is *Scenario 1* (red), and the lowest peak corresponds to *Scenario 2* (green), identify – providing explanations – which curve corresponds to the following scenarios.

- The recovery delay increases from 2 to 2.5
- The contact rate drops from 6 to 5



[4]